

LCOI: Local crystalline orientation identifier using mask fitting implemented in MATLAB

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Statement of need

Detection and analysis of the orientation of local crystalline order arises in different fields such as chemistry, materials science, and condensed matter physics. In particular, such analysis is central for studying crystal melting, jamming, crystal twinning, and crystal strains, among others. Although the local orientations can be deduced from the diffraction data (Angel et al., 2015; Busing & Levy, 1967; Zaefferer, 2011), such analysis must take some spatial averages. Several software packages analyze local crystalline order based solely on real-space imaging data. One widely used approach is the bond-orientational order parameter (BOOP), originally introduced by Steinhardt et al. (Steinhardt et al., 1983), and implemented in various tools (Menon et al., 2019; Ramasubramani et al., 2020; Stukowski, 2009), which provide quantitative tools to classify local environments. These tools are commonly used in both simulations and experimental particle tracking contexts. However, BOOP-based methods have notable limitations, including their reliance on isotropic assumptions, high sensitivity to defects, and a lack of robustness to small spatial fluctuations (Mickel et al., 2013). Some commercial tools also support symmetry and orientation analysis, especially from diffraction data. For example, CrysTBox (Klinger et al., 2016) use Fourier transforms and lattice fitting methods, primarily working in the frequency domain. It offers limited orientation mapping directly from high-resolution transmission electron microscopy images via geometric phase analysis and Fourier-based methods. Our approach departs from both diffraction-based and BOOP-based methods. We propose an approach based on optimally fitting a mask of specific shape to the detected particles. Within this approach, the resulting directionality is continuous and robust to local noise. The accuracy in mask orientation detection is primarily limited by the accuracy of the particles' localization, rather than from the symmetry assumptions or discrete angular jumps typical of BOOP-based metrics.

Summary

LCOI is designed to extract orientation information about local crystalline order from microscopy images such as transmission electron microscopy (TEM). Specifically, the software is able to identify triangular, rectangular and hexagonal order within planar cross-sections of colloidal crystals. The orientation identification is based on mask fitting to the identified particles. The software, implemented as a MATLAB (The MathWorks Inc., 2023) application and available via [GitHub](#), provides tools for basic image processing, particle identification, mask creation and fitting, orientation analysis, and strain measurement in crystals.

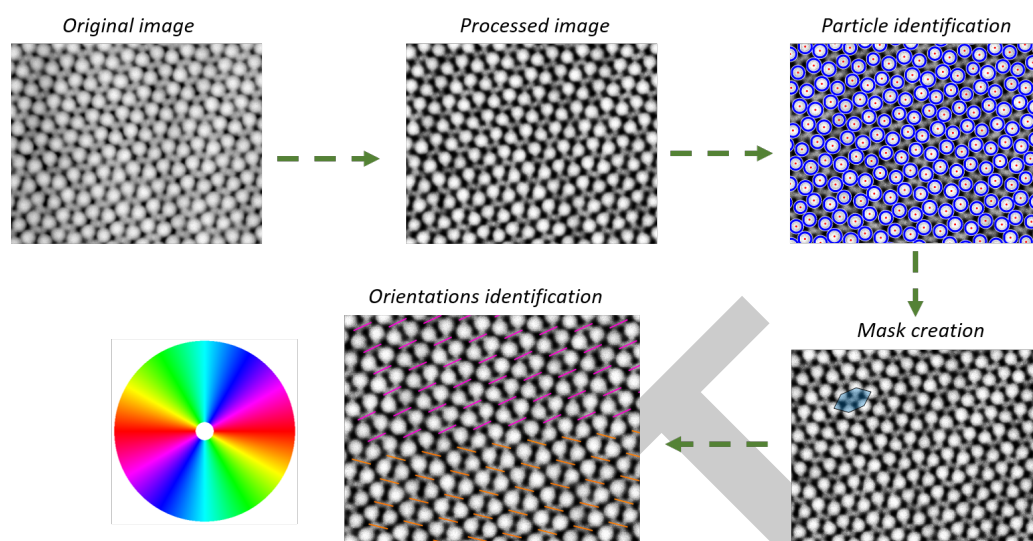


Figure 1: Summary of the general workflow of the orientation identification process, demonstrated using TEM data. The result shows the color-coded orientations of the identified 2-fold symmetric hexagonal crystalline order.

After loading the image and optionally adjusting the contrast, the user can apply particle detection or upload pre-identified data. The user then creates a mask (hull) by manually selecting particles of the desired symmetry, which is subsequently symmetrized by the software. The options are triangular, rectangular and hexagonal, where the last two can either have 2-fold symmetry or 4-fold and 6-fold, respectively. Next, the local orientation of the optimal fits of the created mask to the identified particles is detected and color-coded.

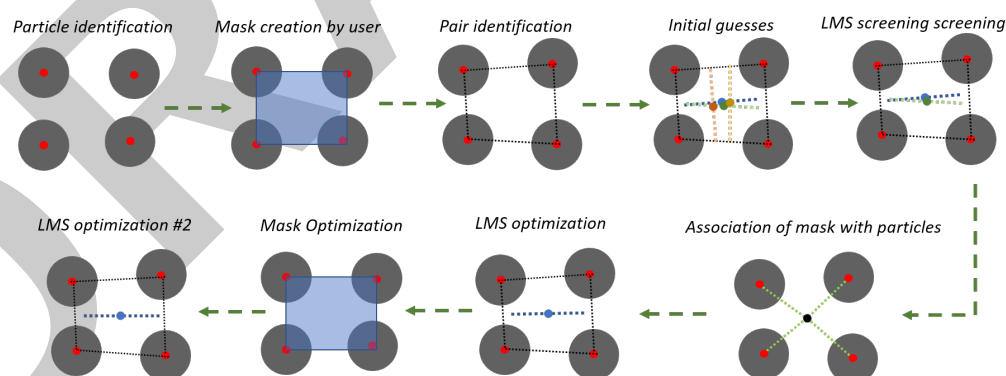


Figure 2: Detailed schematic representation of the process of optimal fitting of the masks.

After particle identification and mask creation, the following step is identification of pairs of particles by considering all particles below a distance threshold set by considering the created mask. Then, a fitting mask is located as if each pair corresponds to an edge of the mask. These initial guesses are screened for the sum of the distances to the closest particles to each mask's vertex. Next, for the screened masks' location and orientation, the particle associated with each mask's vertex is identified and the optimal location and orientation is computed analytically. For more details, see the attached file in the repository: [Mask location and rotation optimization](#). Finally, the mask can be refined using all identified fits by averaging the relevant distances, after which the optimal location and orientation are recomputed.

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References

- Angel, R., Milani, S., Alvaro, M., & Nestola, F. (2015). OrientXplot: A program to analyse and display relative crystal orientations. *Journal of Applied Crystallography*, 48(4), 1330–1334. <https://doi.org/10.1107/S160057671501167X>
- Busing, W. R., & Levy, H. A. (1967). *Angle calculations for 3-and 4-circle x-ray and neutron diffractometers*. Oak Ridge National Lab.(ORNL), Oak Ridge, TN (United States). <https://doi.org/10.1107/S0365110X67000970>
- Klinger, M., Polívka, L., Jäger, A., & Tyunina, M. (2016). Quantitative analysis of structural inhomogeneity in nanomaterials using transmission electron microscopy. *Journal of Applied Crystallography*, 49(3), 762–770. <https://doi.org/10.1107/S1600576716003800>
- Menon, S., Leines, G. D., & Rogal, J. (2019). Pyscal: A python module for structural analysis of atomic environments. *Journal of Open Source Software*, 4(43), 1824. <https://doi.org/10.21105/joss.01824>
- Mickel, W., Kapfer, S. C., Schröder-Turk, G. E., & Mecke, K. (2013). Shortcomings of the bond orientational order parameters for the analysis of disordered particulate matter. *The Journal of Chemical Physics*, 138(4), 044501. <https://doi.org/10.1063/1.4774084>
- Ramasubramani, V., Dice, B. D., Harper, E. S., Spellings, M. P., Anderson, J. A., & Glotzer, S. C. (2020). Freud: A software suite for high throughput analysis of particle simulation data. *Computer Physics Communications*, 254, 107275. <https://doi.org/10.1016/j.cpc.2020.107275>
- Steinhardt, P. J., Nelson, D. R., & Ronchetti, M. (1983). Bond-orientational order in liquids and glasses. *Physical Review B*, 28(2), 784. <https://doi.org/10.1103/PhysRevB.28.784>
- Stukowski, A. (2009). Visualization and analysis of atomistic simulation data with OVITO—the open visualization tool. *Modelling and Simulation in Materials Science and Engineering*, 18(1), 015012. <https://doi.org/10.1088/0965-0393/18/1/015012>
- The MathWorks Inc. (2023). *MATLAB version: 9.14.0 (R2023a)*. The MathWorks Inc. <https://www.mathworks.com>
- Zaefferer, S. (2011). A critical review of orientation microscopy in SEM and TEM. *Crystal Research and Technology*, 46(6), 607–628. <https://doi.org/10.1002/crat.201100125>